

Where and How Fast Is Heat Stress Worsening? A District-Level, Multi-Index Machine-Learning Projection for 64 districts of Bangladesh

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Abstract

Rising heat and humidity increasingly threaten health, labour, and energy systems across Bangladesh, yet operational planning needs district-resolved, uncertainty-aware projections of the specific conditions that drive human heat stress. We present a deterministic, reproducible machine-learning framework that projects seven temperature-correlated heat-stress conditions—heat-index comfort, wet-bulb temperature, indoor and outdoor wet-bulb globe temperature, cooling degree-days, ultraviolet exposure, and dew-point moisture stress—for all 64 districts of Bangladesh. Four gradient-boosting ensembles are trained on the 2014–2024 monthly record using future-known seasonal features; the observed 1980–2024 Theil–Sen trend is superimposed so that projections evolve realistically, and all thermal indices are recomputed from the projected primary variables to avoid target leakage. Validated against withheld 2025 observations, the projected indices attain median R^2 of 0.95–0.98, and their operational risk classes match observations

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to within one class in 100% of district-months. A Mann–Kendall analysis shows maximum temperature rising in every district (median $+0.56^{\circ}\text{C}$ per decade) while minimum temperature remains flat, indicating a widening diurnal range; the coastal and southwestern zones carry the greatest combined burden, and ultraviolet and oppressive-humidity hazards are effectively nationwide. District choropleths, zonal envelopes, and calibrated uncertainty bands provide a directly actionable heat-stress outlook while transparently bounding the models’ skill relative to climatology.

Keywords: heat stress, wet-bulb temperature, WBGT, seasonal projection, gradient boosting, Mann–Kendall trend, Bangladesh, climate adaptation

1. Introduction

Anthropogenic climate change has fundamentally altered the frequency, intensity, and duration of extreme heat events worldwide. Mean surface air temperatures have risen steadily since the pre-industrial era, and this warming has been accompanied by a disproportionate increase in heatwaves, prolonged humid spells, and compound heat–humidity extremes that the human body increasingly struggles to tolerate [1]. Unlike many climate hazards, extreme heat acts as a silent stressor, producing cumulative physiological strain that is frequently underreported in official mortality and morbidity statistics yet carries substantial public-health, occupational, and economic consequences [2]. Understanding how heat stress manifests across different climatic and geographic settings has therefore become an urgent priority for adaptation planning.

Air temperature alone does not represent the heat stress the human body actually experiences and can lead to misleading assessments. Relative

16 humidity, solar radiation, wind, and atmospheric moisture all govern the
17 body’s ability to shed heat through evaporation, radiation, and convection;
18 elevated humidity in particular suppresses the evaporation of sweat—the
19 body’s primary cooling mechanism—so that core temperature rises even
20 when the air is not exceptionally hot [3]. To capture these combined effects,
21 researchers have developed composite heat-stress indices that now underpin
22 public-health warning systems, occupational heat-safety standards [4], and
23 cooling-energy demand projections [5], and that increasingly recognise the
24 approach of physiological survivability limits under extreme humid heat [6].

25 South Asia is among the regions most acutely exposed to escalating heat
26 stress, owing to its dense population, high monsoon-season humidity, and
27 large share of outdoor and informal-sector labour. Bangladesh occupies a
28 uniquely exposed position within it: a tropical monsoon climate, low-lying
29 deltaic geography, and exceptionally high population density combine with
30 rapid, unplanned urbanization to let heat stress intensify quickly and dis-
31 tribute unevenly across the landscape [7]. Critically, the country is not
32 climatically homogeneous—its territory spans drought-prone northwestern
33 plains with continental microclimates, a densely built and heat-retentive
34 central urban core exhibiting strong urban heat-island effects, low-lying
35 coastal districts shaped by maritime humidity, seasonally inundated wetland
36 basins, and elevated hill tracts with distinct topographic regimes. Each zone
37 is expected to experience thermal stress through different pathways and at
38 different intensities, yet most existing heat-risk assessments in Bangladesh
39 continue to treat the country as a single climatic unit or rely on a narrow
40 selection of indices—most commonly ambient temperature alone—without
41 accounting for the multi-dimensional nature of heat exposure [8].

42 A parallel gap exists in temporal scope and forecasting capability. Most

43 prior work on heat stress in Bangladesh is retrospective in orientation or
 44 resolved only to divisional or national summaries, and very few studies in-
 45 tegrate long-term climatological baselines with near-term machine-learning
 46 forecasting to project how thermal stress will evolve at the district level [**verify-**
 47 **intended**]. This omission matters, because heat-risk management—whether
 48 through early-warning systems, occupational health policy, or urban cooling
 49 infrastructure—requires knowing not only where risk is highest today but
 50 where and how quickly it is worsening. A recent wet-bulb globe tempera-
 51 ture analysis across Bangladesh reported warming of roughly 0.08–0.5 °C per
 52 decade, most pronounced in the southeast and northeast [9], underscoring
 53 the need for district-resolved, multi-index projection.

54 This study addresses these gaps through the Temperature-Related
 55 Conditions (TRC) framework: an integrated set of seven temperature-dependent
 56 indicators —heat-index comfort, wet-bulb temperature, indoor and outdoor
 57 wet-bulb globe temperature, heatstroke risk, cooling degree-days, ultravi-
 58 olet exposure, and dew-point moisture stress—each capturing a distinct
 59 physiological, biophysical, or infrastructural dimension of heat-related risk.
 60 Applying this framework to 45 years (1980–2024) of district-level meteo-
 61 rological records across all 64 districts and six climatically distinct zones,
 62 we (i) train an ensemble of gradient-boosting models [10] on future-known
 63 seasonal features and re-impose the observed long-term trend so that projec-
 64 tions evolve realistically rather than collapsing onto the last training year;
 65 (ii) recompute every thermal index from the projected primary variables
 66 rather than predicting the indices directly, eliminating target leakage and
 67 preserving the physical coupling among temperature, humidity, and radi-
 68 ation; (iii) generate 24-month district-level projections (2025–2026) with
 69 analytically propagated uncertainty bands; and (iv) validate the projections

70 out-of-sample against independent 2025 observations, benchmark their skill
71 transparently against seasonal-naive and climatological baselines, and char-
72 acterise long-term change with the non-parametric Mann–Kendall test. The
73 best model per variable is selected by a multi-attribute simple-additive-
74 weighting scheme with a generalization-gap penalty [11].

75 Two features distinguish the analysis. First, results are reported at full
76 district resolution through choropleth maps rather than aggregated away,
77 exposing spatial gradients—such as the coastal concentration of humid-
78 heat burden—that zonal summaries obscure. Second, we are explicit about
79 the limits of predictive skill: at monthly resolution the ensembles essen-
80 tially match climatology, so their value lies in providing spatially complete,
81 uncertainty-aware, and trend-consistent projections of physiologically rele-
82 vant indices rather than in reducing monthly error. One indicator, heat-
83 stroke risk, was found to lack out-of-sample skill and is retained only de-
84 scriptively. The remainder of this paper details the data and methods,
85 presents the validated projections and observed trends, and discusses their
86 implications for heat-adaptation planning across Bangladesh.

87 **2. Literature Review**

88 Human exposure to heat is not adequately captured by dry-bulb air tem-
89 perature alone. A growing body of work therefore evaluates compound ther-
90 mal indices that combine temperature with humidity, radiation, and wind
91 to describe how heat is actually experienced and how it threatens health,
92 labour productivity, and energy demand. This section synthesises recent in-
93 ternational and Bangladesh-focused literature across the seven temperature-
94 correlated conditions addressed in this study, the machine-learning methods

95 used to forecast them, and the urban and regional context that motivates a
96 district-resolved, multi-index treatment.

97 *2.1. The physiological basis of thermal-stress indices*

98 Raymond et al. [6] provided seminal evidence that a wet-bulb tempera-
99 ture (WBT) of 35 °C represents the physiological limit for human survival,
100 showing that humid-heat extremes had already more than doubled in fre-
101 quency between 1979 and 2017 and shifting the field’s focus from ambient
102 temperature toward compound heat–humidity metrics. The inadequacy of
103 dry-bulb temperature as a standalone indicator is well established: high hu-
104 midity impairs the evaporative efficiency of perspiration, causing perceived
105 loads to exceed measured air temperature [3]. The heat index formalises this
106 through the regression of Rothfus [12], operationalised by the U.S. National
107 Weather Service. The wet-bulb globe temperature (WBGT), originally de-
108 veloped by Yaglou and Minard [13] for military training and codified in
109 ISO 7243 [4], extends the framework to radiant load, making it the interna-
110 tional standard for occupational heat-stress assessment [14]. The empirical
111 WBT approximation of Stull [15], derived by gene-expression programming
112 with a mean absolute error below 0.3 °C, provides a computationally efficient
113 estimate applicable to station and reanalysis data alike. Cooling degree-days
114 (CDD) complement these physiological indices by quantifying the cumula-
115 tive thermal energy above a baseline, a standard proxy for building cooling
116 demand [5, 16]; the UV index [17] captures the solar-radiation dimension of
117 outdoor exposure; and dew-point temperature offers a stable, temperature-
118 independent measure of atmospheric moisture that outperforms relative hu-
119 midity as a predictor of perceived discomfort [18]. Together these conditions
120 span the physiological, biophysical, and infrastructural dimensions of heat

121 risk that the present study projects jointly, against a backdrop of accelerat-
122 ing global warming documented by the IPCC [19].

123 *2.2. Observed heat-stress trends in Bangladesh*

124 Bangladesh is among the most heat-exposed nations, and a dense body
125 of station-based work documents its warming. Rahman et al. [20] pro-
126 duced the first comprehensive spatiotemporal appraisal of historical and
127 projected heat-index change, finding statistically significant upward trends
128 across most districts, sharpest in the pre-monsoon season. Ekra et al. [21]
129 analysed 60 years (1961–2020) of records from 17 stations using the Discom-
130 fort Index, Humidex, and WBT, reporting a spatially uneven rise in heat
131 discomfort that is steepest along the coastal belt and driven primarily by
132 rising air temperature. For WBGT specifically, Kamal et al. [9] provided
133 the anchor analysis, computing outdoor WBGT from 1979–2021 reanaly-
134 sis with the physically based Liljegren model and finding a national rise of
135 0.08–0.5 °C per decade, a fivefold expansion of the area affected by high-to-
136 extreme stress, and air temperature as the dominant driver (85.9% relative
137 importance); a companion paper developed simplified regression equations
138 to lower the computational barrier to WBGT adoption [22]. Temperature-
139 extreme indices are likewise intensifying: Ahmed et al. [23] found length-
140 ening warm-spell durations, particularly in coastal regions, and Islam et al.
141 [24] identified warm-index trends dominated by post-2000 increases across
142 27 stations. Trends in degree-days were examined by Islam et al. [25], who
143 reported significant CDD increases with the southwestern and central zones
144 showing the highest annual means. At the health end, Miah et al. [26] linked
145 ambient temperature and heat index to heatstroke mortality in an ecological
146 time-series analysis, reinforcing the public-health urgency of a multi-index

147 approach.

148 *2.3. Urban heat islands and zone-specific dynamics*

149 Urban heat-island (UHI) intensity amplifies background warming, espe-
150 cially within the densely built central zone examined here. Dewan et al.
151 [27] characterised diurnal and seasonal UHI across five major Bangladeshi
152 cities from satellite land-surface temperature, finding the greatest magni-
153 tude in the pre-monsoon period and the largest daytime surface UHI in
154 Dhaka (2.74 °C), followed by Chittagong (1.92 °C) and Khulna (1.27 °C).
155 Faisal et al. [28] reported that UHI intensity rose substantially across most
156 districts from 2000 to 2020, with Dhaka and Mymensingh steepest, and
157 Park et al. [29] showed that Dhaka’s UHI is strongest in winter (0.95 °C)
158 and weakest during the monsoon (0.23 °C), with heat waves producing dis-
159 tinctive synoptic patterns that compound local warming. These findings
160 support treating Bangladesh’s urban core as a distinct climatic zone and,
161 more broadly, motivate the six-zone structure adopted in this study.

162 *2.4. Machine learning for meteorological and thermal-index forecasting*

163 Ensemble tree methods have consistently outperformed conventional sta-
164 tistical and numerical approaches for temperature and humidity forecast-
165 ing in data-rich, strongly seasonal settings. The gradient-boosting family—
166 XGBoost [30], LightGBM [31], and CatBoost [32]—recurs as a strong, inter-
167 pretable baseline across environmental prediction problems. In Bangladesh,
168 Islam et al. [33] found XGBoost most accurate for seasonal temperature
169 prediction across 29 stations ($R^2 = 0.88$ for maximum and 0.97 for mini-
170 mum temperature) when trained on ENSO predictors; Akter and Rahman
171 [34] found Random Forest best for humidity forecasting at northern sta-

172 tions over a 40-year record; and Rahman et al. [35] found a stacking en-
 173 semble outperformed RNN-LSTM and decision-forest models for average-
 174 temperature prediction over a 60-year series. Most directly, Binte Ahmed
 175 et al. [36] benchmarked ARIMAX, SARIMAX, Random Forest, XGBoost,
 176 and LSTM on 3-hourly Dhaka records (2014–2023) for conditional, scenario-
 177 based heat-index projection, with Random Forest achieving the lowest error
 178 (RMSE = 0.85 °C, $R^2 = 0.987$)—a close methodological precedent for the
 179 present work. The value of combining gradient-boosting learners is illus-
 180 trated by Kim et al. [37], whose stacked XGBoost/LightGBM/CatBoost
 181 ensemble reached $R^2 = 0.93$, outperforming any single model. Fourier-
 182 based encoding of seasonal cycles as sine and cosine terms has been shown
 183 to improve both accuracy and efficiency on meteorological series [38], di-
 184 rectly motivating the Fourier feature layer used here. Comparative studies
 185 elsewhere reinforce the transferability of these methods [39, 40], while the
 186 validation of deep-learning weather models on recent extreme events [41]
 187 and analyses of the distinct atmospheric drivers of humidex versus temper-
 188 ature heatwaves [42] situate the forecasting problem in its broader physical
 189 context.

190 2.5. *Per-condition modelling and international context*

191 Beyond the anchor WBGT and heat-index studies, the individual con-
 192 ditions have distinct literatures. For the heat-index/Humidex family, So-
 193 jan and Srinivasan [43] showed that humid-heat stress in India is intensify-
 194 ing disproportionately faster than dry heat; Kushwaha et al. [44] projected
 195 air temperature as the dominant driver of future extreme-Humidex days
 196 across 15 CMIP6 models; Scoccimarro et al. [45] established that perceived-
 197 temperature extremes intensify faster than temperature extremes owing to

198 rising humidity; Saha et al. [46] documented a post-2000 escalation of humid
 199 heatwaves across Northeast India and Bangladesh; and Tabassum et al. [47]
 200 connected a cool-roof intervention to measurable Humidex reductions during
 201 a Dhaka heat wave. Heatstroke-risk modelling has matured chiefly outside
 202 Bangladesh—Xu et al. [48] trained interpretable models linking meteorol-
 203 ogy to heatstroke incidence in South China—with Ehsan et al. [49] provid-
 204 ing a recent Bangladesh-focused machine-learning forecast of extreme-heat
 205 health impact; consistent with this thin evidence base, the present study
 206 computes a heatstroke indicator but, as reported below, excludes it from
 207 the validated results for a lack of out-of-sample skill. For CDD, Bilgili
 208 [50] showed that classical SARIMA models remain competitive with ma-
 209 chine learning when the seasonal structure is strong. UV-index modelling
 210 is comparatively underexplored in the regional literature: Teggi et al. [51]
 211 recently derived the UV index from reanalysis validated against ground sta-
 212 tions, whereas in the present framework the UV index is instead predicted
 213 directly as a modelled variable. Dew-point prediction has a well-developed
 214 machine-learning lineage, progressing from extreme-learning-machine and
 215 kernel baselines [52, 53], to gradient-boosted trees exploiting cross-station in-
 216 formation [54], to attention-based sequence models with explainability [55].
 217 Internationally, the stakes are large: Parsons et al. [56] estimated humid-
 218 heat labour losses equivalent to roughly 1.7% of 2017 global GDP, with
 219 South Asian outdoor workers among the worst affected, and Ioannou et al.
 220 [57] meta-analysed a significant negative association between WBGT expo-
 221 sure and manual work capacity. Longer-term projections for Bangladesh un-
 222 der CMIP6 [58] indicate continued rises in warm-spell duration and tropical
 223 nights, providing the multi-decadal backdrop against which the near-term
 224 (2025–2026) forecasts generated here should be read.

225 2.6. Research gaps addressed by this study

226 Three gaps emerge from this review. First, although individual indices—
227 chiefly WBGT and the heat index—have been examined for Bangladesh in
228 isolation, no study has simultaneously computed, validated, and forecast
229 seven complementary thermal-stress conditions within a single framework,
230 at district resolution, across climatically distinct zones. Second, existing
231 Bangladeshi forecasting work has relied on conventional statistics, CMIP6
232 downscaling, or single-algorithm machine learning; the multi-algorithm gradient-
233 boosting ensemble with Fourier feature engineering and an explicitly re-
234 imposed observational trend proposed here has not previously been applied
235 to thermal-stress forecasting in this context. Third, the six-zone climato-
236 logical structure of Bangladesh—Northwest dry-hot plains, Central urban
237 core, Southwest saline-drought belt, Coastal maritime zone, Haor wetland
238 basins, and Hill Tracts—has not previously organised a multi-index heat-
239 risk assessment, leaving intra-national thermal heterogeneity systematically
240 undercharacterised. In contrast to studies built on ERA5 reanalysis, the
241 present work is driven by station-derived Visual Crossing records and val-
242 idated out-of-sample against independent 2025 observations, providing an
243 externally verified, uncertainty-aware, and district-resolved projection across
244 all seven conditions.

245 3. Materials and Methods

246 This study builds a deterministic, reproducible pipeline that (i) learns
247 the monthly climatology of seven primary meteorological variables for each
248 of the 64 districts of Bangladesh from a recent, high-quality window; (ii) su-
249 perimposes the long-term observed climate trend so that projections evolve

250 realistically beyond the training period; (iii) recomputes physiologically rel-
 251 evant thermal indices from the projected primary variables to avoid target
 252 leakage; (iv) quantifies projection uncertainty by first-order error propaga-
 253 tion; and (v) validates the projected 2025 fields against independent obser-
 254 vations. All methodological constants are fixed a priori and are reported
 255 here so that the workflow can be audited and reproduced exactly.

256 3.1. Study area and climatic zonation

257 The analysis covers all 64 administrative districts of Bangladesh. For
 258 synthesis and reporting, districts are grouped into six climatic zones: (1) North-
 259 west Extreme Heat Zone (dry hot plains, including the Barind Tract), (2) Cen-
 260 tral Urban Heat Zone (urban heat-island region), (3) Southwest Hot–Dry
 261 Zone (drought- and salinity-affected areas), (4) Coastal Humid Heat Zone
 262 (the coastal belt), (5) Haor/Wetland Humid Zone (the haor basin), and
 263 (6) Hill Tract Zone (elevated terrain). The zonation is author-defined but
 264 grounded in converging, established frameworks: the agro-climatic regionali-
 265 sation of the Bangladesh Meteorological Department and the agro-ecological/physiographic
 266 zonation of Bangladesh [59, 60], together with recognised hydro-ecological
 267 units (Barind Tract, Sundarbans coastal belt, haor basin). Each district
 268 is flagged as a *confirmed* (high-confidence) or *provisional* assignment, and
 269 one representative *exemplar* district per zone is retained for main-text fig-
 270 ures (Figure 1). The complete district-to-zone mapping is provided in the
 271 supplementary material.

272 3.2. Data and preprocessing

273 Daily meteorological records were obtained from the Visual Crossing
 274 Weather API [61] for each district. Seven primary (physically measured)

Climatic zonation of Bangladesh (64 districts, 6 zones)

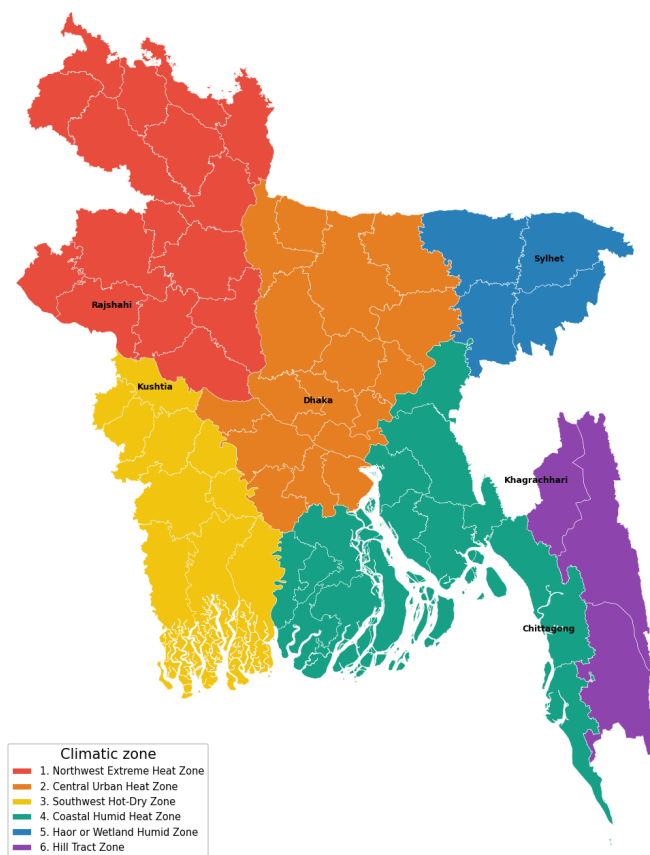


Figure 1: Climatic zonation of Bangladesh: the 64 districts grouped into six zones, with the exemplar district of each zone labelled.

variables are used: mean, maximum, and minimum near-surface air temperature ($^{\circ}\text{C}$); relative humidity (%); dew-point temperature ($^{\circ}\text{C}$); downward solar (shortwave) radiation; and the ultraviolet (UV) index. The historical record spans 1980–2024; an additional file covering 2025 (through 20 November 2025; the final month is partial) is reserved exclusively for out-of-sample validation.

Because humidity is noisy in the earliest decades and solar/UV are unavailable before 2014, the machine-learning window is restricted to 2014–2024, whereas the full 1980–2024 record is used only for the observational trend analysis (Section 3.12). Daily values are aggregated to calendar-month means (month-start resampling), and residual gaps are filled by linear interpolation. All modelling is performed on these monthly series.

3.3. Feature construction

To respect a strict *future-known-features-only* constraint—no lags, rolling windows, or autoregressive terms that would be unavailable at projection time—each month is described by a deterministic seasonal–secular feature vector. Seasonality is encoded by a truncated Fourier series with period $P = 12$ months and $H = 3$ harmonics,

$$\sin_h(m) = \sin\left(\frac{2\pi hm}{P}\right), \quad \cos_h(m) = \cos\left(\frac{2\pi hm}{P}\right), \quad h = 1, \dots, H, \quad (1)$$

where m is the calendar-month index. The secular component is the calendar year. The full design vector is therefore $\mathbf{x} = [\text{year}, \sin_1, \cos_1, \dots, \sin_3, \cos_3]$ (seven features).

3.4. Predictive models

Four gradient-based tree ensembles are trained independently for every district–variable pair: Random Forest [10], XGBoost [30], LightGBM [31],

294 and CatBoost [32]. Hyperparameters are fixed a priori and applied uniformly
 295 across all districts and variables so that results are mutually comparable
 296 and the pipeline is fully deterministic (global random seed = 42). The fixed
 297 configurations are listed in the supplementary material.

298 3.5. *Trend-aware detrending and projection*

Tree ensembles cannot extrapolate beyond the range of their training features; if the calendar year is used directly as a predictor, every projected year collapses onto the last training year, producing a static (trendless) forecast. To carry the observed climate trend forward while preserving the future-known constraint, each primary variable is detrended before learning and re-trended at prediction time. For district-variable series $y(t)$ with continuous time t (in years), a Theil-Sen slope $\hat{\beta}$ is estimated from the annual-mean series,

$$\hat{\beta} = \text{median}_{i < j} \frac{\bar{y}_j - \bar{y}_i}{t_j - t_i}, \quad (2)$$

299 where $\bar{y}_k$ is the mean of year k . To keep the projected trend consistent with
 300 the manuscript’s observational trend analysis, $\hat{\beta}$ is taken from the robust
 301 1980–2024 record for the five variables that span it (mean/max/min tem-
 302 perature, humidity, dew point); solar radiation and the UV index, which
 303 lack a pre-2014 record, fall back to a Theil-Sen fit over their available 2014–
 304 2024 window. The ensembles are trained on the deseasonalised residual
 305 $r(t) = y(t) - \hat{\beta}t$, so they model only the seasonal signal; the linear trend
 306 is added back at prediction time, $\hat{y}(t) = \hat{r}(t) + \hat{\beta}t$. This design makes each
 307 projected year differ from the previous one by exactly $\hat{\beta}$ and guarantees that
 308 the projected inter-annual tendency equals the reported Mann-Kendall/Sen
 309 slope (Section 3.12). We emphasise that this trend term is included for
 310 physical consistency with the observed record, not to improve short-horizon

accuracy: at monthly resolution the seasonal cycle dominates, so the trend contributes only a small secular adjustment.

The winning model per variable (Section 3.6) is refit on the full 2014–2024 residual and projected forward for a 24-month horizon (January 2025 through December 2026). Projected primary variables are clipped to physically plausible bounds before any downstream computation to prevent non-physical extrapolations from propagating into the indices.

3.6. Model evaluation and selection

Reported test metrics use a strictly chronological hold-out: models are trained on 2014–2022 and tested on the withheld 2023–2024 period. Generalisation is additionally assessed by five-fold `TimeSeriesSplit` cross-validation on the residual series. For each candidate model we record the coefficient of determination R^2 and root-mean-square error (RMSE) on the chronological test set, the cross-validated R^2 ($CV\text{-}R^2$), a generalisation gap $\text{Gap} = |R^2 - CV\text{-}R^2|$, and an operational tolerance accuracy A_τ defined as the fraction of test months whose absolute error does not exceed a variable-specific tolerance τ ,

$$A_\tau = \frac{1}{N} \sum_{n=1}^N \mathbf{1}(|y_n - \hat{y}_n| \leq \tau). \quad (3)$$

The winning model is chosen by Simple Additive Weighting (SAW), a multi-attribute decision-making scheme [11], with a generalisation-gap penalty. Each criterion c is min–max normalised across the four candidates—directly for benefit criteria and by complement for cost criteria,

$$\tilde{s}_c = \begin{cases} \frac{s_c - \min_c}{\max_c - \min_c}, & c \text{ maximised,} \\ \frac{\max_c - s_c}{\max_c - \min_c}, & c \text{ minimised,} \end{cases} \quad (4)$$

319 and the composite score is the weighted sum $S = \sum_c w_c \tilde{s}_c$, with weights
 320 $w_{R^2} = 0.25$, $w_{CV-R^2} = 0.25$, $w_{RMSE} = 0.20$, $w_{Gap} = 0.20$, and $w_{A_\tau} = 0.10$
 321 (RMSE and Gap treated as cost criteria). The highest-scoring model wins;
 322 models with Gap > 0.10 are additionally flagged as overfitting risks.

323 3.7. Thermal index computation

To avoid target leakage, thermal-stress indices are never predicted directly; they are recomputed analytically from the *projected* primary variables. The heat index (HI) uses the U.S. National Weather Service Rothfus regression [12]. With temperature T_F and relative humidity R (in °F and %),

$$\begin{aligned} \text{HI} = & -42.379 + 2.04901523 T_F + 10.14333127 R - 0.22475541 T_F R \\ & - 6.83783 \times 10^{-3} T_F^2 - 5.481717 \times 10^{-2} R^2 + 1.22874 \times 10^{-3} T_F^2 R \\ & + 8.5282 \times 10^{-4} T_F R^2 - 1.99 \times 10^{-6} T_F^2 R^2, \end{aligned} \quad (5)$$

with the standard low-humidity and high-humidity adjustments applied within their validity ranges and a simplified formula substituted below 80°F; the result is returned in °C. Wet-bulb temperature T_w follows the empirical formulation of Stull [15],

$$\begin{aligned} T_w = & T \arctan[0.151977 (R + 8.313659)^{1/2}] + \arctan(T + R) \\ & - \arctan(R - 1.676331) + 0.00391838 R^{3/2} \arctan(0.023101 R) - 4.686035, \end{aligned} \quad (6)$$

with T in °C and R in %. Wet-bulb globe temperature is computed in a shade variant (no direct-beam term) and a sun variant that adds a solar-

radiation contribution S ,

$$\text{WBGT}_{\text{shade}} = 0.7 T_w + 0.3 T, \quad (7)$$

$$\text{WBGT}_{\text{sun}} = 0.7 T_w + 0.2 (T + 0.025 S) + 0.1 T. \quad (8)$$

Cooling degree-days are expressed as a monthly-mean daily rate about an 18°C base,

$$\text{CDD} = \max\left(\frac{T_{\text{max}} + T_{\text{min}}}{2} - 18, 0\right). \quad (9)$$

324 A composite heatstroke-hazard score (a sigmoid of the heat index mod-
 325 ulated by a wet-bulb multiplier) was also implemented and recalibrated for
 326 monthly-mean inputs. However, in external validation it produced a nega-
 327 tive out-of-sample R^2 (i.e. worse than a constant predictor), so it is *excluded*
 328 *from all predictive results* and retained only as a descriptive projected field.
 329 This exclusion is stated explicitly to avoid over-interpretation of an index
 330 that does not demonstrate skill.

331 3.8. Operational heat-stress conditions and risk categorisation

332 The projected fields are organised into seven temperature-correlated heat-
 333 stress conditions of operational relevance: (1) heat-index comfort (perceived
 334 temperature under humidity); (2) wet-bulb temperature (the thermody-
 335 namic limit of evaporative cooling); (3) wet-bulb globe temperature in in-
 336 door (shade) and outdoor (sun) forms (the occupational heat-exposure stan-
 337 dard); (4) heatstroke risk; (5) cooling degree-days (a building-cooling energy
 338 proxy); (6) ultraviolet exposure risk; and (7) dew-point moisture stress.
 339 Conditions (6) and (7) are obtained directly from the projected UV-index
 340 and dew-point fields, which are among the primary modelled variables (Sec-
 341 tion 3.2); condition (4) is excluded from predictive results as noted above.

342 To move from continuous values to operational meaning, four of the
 343 conditions are additionally mapped to established risk-category systems:
 344 the U.S. National Weather Service heat-index comfort classes (Comfortable,
 345 Caution, Extreme Caution, Danger, Extreme Danger); WBGT occupational-
 346 exposure classes (Low, Moderate, High, Extreme); the World Health Orga-
 347 nization UV-index classes (Low, Moderate, High, Very High, Extreme) [17];
 348 and dew-point human-comfort classes (Dry through Extremely Oppressive).
 349 Wet-bulb temperature (2) and cooling degree-days (5) are reported as con-
 350 tinuous physical quantities. The category boundaries are conventionally de-
 351 fined for daily peak values; because the present pipeline operates on monthly
 352 means, the extreme classes are seldom reached and the categories should be
 353 read as describing *typical monthly* conditions rather than daily peaks. Cat-
 354 egorisation skill is assessed against the observed 2025 fields by exact-match
 355 agreement and by agreement to within one adjacent class.

356 3.9. Uncertainty quantification

Projection uncertainty for the temperature–humidity indices is obtained
 by first-order (linearised) error propagation from the primary-variable un-
 certainties. For an index $I(x_1, \dots, x_k)$ with input standard deviations σ_{x_j} ,

$$\sigma_I = \left[\sum_j \left(\frac{\partial I}{\partial x_j} \right)^2 \sigma_{x_j}^2 \right]^{1/2}, \quad (10)$$

357 where the input σ_{x_j} are taken from each winning model’s in-sample RMSE
 358 and representative sensitivity coefficients are used for the heat index, wet-
 359 bulb temperature, and WBGT. The propagated σ_I define the 68% ($\pm\sigma_I$)
 360 and 95% ($\pm 1.96\sigma_I$) bands reported for the projected indices.

361 *3.10. Baseline skill benchmarks*

Model skill is benchmarked against two references evaluated on the same chronological test set: a seasonal-naive predictor (each test month set to the same calendar month one year earlier) and a monthly-climatology predictor (the training-period mean for each calendar month). Skill relative to a reference is

$$SS = 1 - \frac{RMSE_{\text{model}}}{RMSE_{\text{ref}}}, \quad (11)$$

362 so that positive values indicate improvement over the reference.

363 *3.11. External validation against observed 2025*

364 The 24-month projection includes calendar year 2025, for which indepen-
 365 dent observations were withheld from all training. Projected monthly pri-
 366 mary variables and recomputed indices are compared with the observed 2025
 367 monthly series using R^2 , RMSE, mean absolute error, and the tolerance-
 368 accuracy bands of Section 3.6. The final month (November 2025) is partial
 369 in the source data and is treated accordingly. This constitutes a genuine,
 370 fully out-of-sample test of the projection.

371 *3.12. Observational trend analysis*

Long-term trends are characterised on the full 1980–2024 annual-mean series using the non-parametric Mann–Kendall test with Theil–Sen slope estimation, computed independently of the predictive models (and therefore free of target leakage). The Mann–Kendall statistic is

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(x_j - x_k), \quad (12)$$

with tie-corrected variance

$$\text{Var}(S) = \frac{n(n-1)(2n+5) - \sum_i t_i(t_i-1)(2t_i+5)}{18}, \quad (13)$$

where t_i is the number of ties of extent i . The standardised statistic

$$Z = \begin{cases} (S - 1)/\sqrt{\text{Var}(S)}, & S > 0, \\ 0, & S = 0, \\ (S + 1)/\sqrt{\text{Var}(S)}, & S < 0, \end{cases} \quad (14)$$

is referred to the standard normal distribution for a two-sided p -value, and Kendall's $\tau = 2S/[n(n - 1)]$ measures rank correlation with time. The Theil–Sen slope (Eq. 2) provides the trend magnitude, reported per decade. Trends with $p < 0.05$ are deemed significant. This analysis is performed for mean, maximum, and minimum temperature, humidity, dew point, and the temperature–humidity indices (heat index, wet-bulb temperature, shade WBGT) that are computable across the full record.

A subset of districts encode pre-~2000 missing days as zero in the source record; left uncorrected, the resulting $0 \rightarrow \sim 30^\circ\text{C}$ step fabricates physically impossible trends (an artefactual $+7.8^\circ\text{C}$ per decade in one district). Before any trend is fit, daily values below physically plausible floors (5°C for temperatures, with lower floors for minimum temperature and dew point, and 5% for humidity) are therefore treated as missing, and a year must retain at least 60 valid days to enter the annual-mean series. The same masked slopes are used when the trend is re-added to the projection (Section 3.5). This affects only the observational trend and long-run slope; the machine-learning models are unaffected because they train exclusively on the complete 2014–2024 window.

3.13. Zonal synthesis

District-level projections are summarised by zone as monthly and annual envelopes (the minimum, mean, and maximum across member districts) for

the key indices, providing compact, journal-page-efficient views of the projected heat-stress distribution in place of 64 individual district panels. Full district-level detail is additionally retained through choropleth maps rendered directly from the CC BY 3.0 IGO geoBoundaries Bangladesh ADM2 administrative boundaries [62]; districts are matched to the boundary geometry by name, and per-district values are mapped for the observed trends, the projected annual means of all seven conditions, and the number of months per year each district spends in each condition’s hazard class.

3.14. Implementation and reproducibility

The pipeline is implemented in Python 3.14 using scikit-learn 1.8 [63], LightGBM 4.6, XGBoost 3.3, CatBoost 1.2.10, together with NumPy, pandas, and Matplotlib. All stochastic components use a fixed seed, and a run manifest records package versions, configuration constants, and per-run timings so that every reported artefact is traceable to a specific configuration.

4. Results

The full pipeline was executed for all 64 districts, producing 448 district-variable models, a 24-month projection (January 2025–December 2026) per district, an out-of-sample validation against observed 2025, and the 1980–2024 observational trend analysis. Unless stated otherwise, statistics are reported as the median across the 64 districts.

4.1. Predictive accuracy and model selection

On the chronological 2023–2024 hold-out, the ensembles reproduced the primary variables accurately, with median test R^2 of 0.970 for minimum temperature, 0.961 for mean temperature, 0.962 for dew point, and 0.886

417 for maximum temperature; humidity (0.632), solar radiation (0.691), and the
 418 UV index (0.664) were intrinsically harder and scored lower. Under the SAW
 419 criterion, LightGBM was selected most frequently (284 of 448 fits), followed
 420 by Random Forest (111) and CatBoost (53); XGBoost, although included in
 421 the candidate pool, was never top-ranked. Model preference was variable-
 422 dependent (Table 1): LightGBM dominated temperature (57/64), maximum
 423 temperature (63/64), and dew point (48/64), whereas Random Forest was
 424 preferred for the noisier humidity field (55/64) and was competitive for solar
 425 radiation and UV.

Table 1: Best-model selection frequency across the 64 districts, by variable (SAW criterion). XGBoost was never selected and is omitted.

Variable	LightGBM	Random Forest	CatBoost
Mean temperature	57	0	7
Maximum temperature	63	0	1
Minimum temperature	44	2	18
Relative humidity	4	55	5
Dew point	48	0	16
Solar radiation	29	29	6
UV index	39	25	0

426 4.2. External validation against observed 2025

427 The projected 2025 fields, produced without any exposure to 2025 ob-
 428 servations, validated strongly (Table 2). Median R^2 reached 0.982 for min-
 429 imum temperature, 0.980 for dew point, and 0.950 for mean temperature
 430 (RMSE 0.72 °C). Crucially, the thermal-stress indices—recomputed from

the projected primary variables rather than predicted directly—also validated well: wet-bulb temperature and shade WBGT both at $R^2 = 0.969$ (RMSE $\leq 0.63^\circ\text{C}$), sun WBGT at 0.966, the heat index at 0.951 (RMSE 1.36°C), and cooling degree-days at 0.951. Maximum temperature (0.845), humidity (0.825), UV (0.805), and solar radiation (0.766) were the weakest, consistent with their lower learnability. For the six exemplar districts, heat-index validation R^2 ranged from 0.873 (Sylhet) and 0.891 (Chittagong) to 0.948–0.954 (Kushtia, Khagrachhari, Rajshahi, Dhaka), with RMSE between 1.1 and 1.9°C .

Table 2: External validation of the projected 2025 fields against observed 2025, summarised as the median across 64 districts. Errors are in each variable’s native unit ($^\circ\text{C}$ for temperatures and thermal indices; % for humidity; W m^{-2} for solar radiation; index units for UV; $^\circ\text{C d d}^{-1}$ for CDD).

Variable	n	Median R^2	Median RMSE	Median MAE
Mean temperature	64	0.950	0.720	0.615
Maximum temperature	64	0.845	0.940	0.788
Minimum temperature	64	0.982	0.565	0.449
Relative humidity	64	0.825	2.334	1.996
Dew point	64	0.980	0.601	0.469
Solar radiation	64	0.766	14.352	11.704
UV index	64	0.805	0.375	0.302
Heat index	64	0.951	1.357	1.142
Wet-bulb temperature	64	0.969	0.624	0.509
WBGT (shade)	64	0.969	0.595	0.506
WBGT (sun)	64	0.966	0.644	0.543
Cooling degree-days	64	0.951	0.696	0.577

4.3. Skill relative to baselines

Against the seasonal-naive reference the ensembles were skilful in 81.5% of district-variable cases, with the largest median gains for minimum temperature (+0.24) and dew point (+0.21). Relative to monthly climatology, however, the models were essentially at parity: the median skill score was near zero or slightly negative for every variable (e.g. -0.003 for mean temperature, -0.030 for maximum temperature, -0.059 for minimum temperature), and only 31.9% of cases exceeded climatology. This is the expected behaviour at monthly resolution: the ensembles reconstruct the strong tropical seasonal cycle that monthly climatology already encodes, so they add little in raw error terms. Their contribution is therefore not a reduction in monthly RMSE but a spatially complete, uncertainty-quantified, and trend-consistent projection that climatology cannot provide.

4.4. Observed long-term trends, 1980–2024

The Mann–Kendall/Sen analysis revealed a coherent warming signal (Table 3, Figure 2). Maximum temperature increased significantly in all 64 districts (median $+0.56^\circ\text{C}$ per decade), and the heat index, wet-bulb temperature, and shade WBGT increased in 58, 57, and 58 of 64 districts (median $+0.37$, $+0.17$, and $+0.17^\circ\text{C}$ per decade, respectively). Mean temperature rose in 50 districts and dew point in 46. Minimum temperature was markedly more heterogeneous (16 increasing, 13 decreasing, 35 without significant trend; median $+0.02^\circ\text{C}$ per decade), and humidity was mixed (24 increasing, 23 decreasing). The sharp contrast between near-universal maximum-temperature warming and the essentially flat minimum-temperature signal indicates a widening diurnal temperature range in much of the country; a small number of districts (e.g. Sylhet) therefore show a slight decrease in

466 *mean* temperature despite rising maxima. Because the projection re-imposes
 467 exactly these Sen slopes, the projected inter-annual tendency reproduces this
 468 observed signal by construction.

Table 3: Observed 1980–2024 trends (Mann–Kendall test with Theil–Sen slope) by variable: number of districts increasing, decreasing, or without a significant trend ($p < 0.05$), and the median Sen slope per decade ($^{\circ}\text{C}$ per decade, except humidity in % per decade).

Variable	Increasing	Decreasing	No trend	Median slope/decade
Maximum temperature	64	0	0	+0.558
Mean temperature	50	2	12	+0.194
Minimum temperature	16	13	35	+0.021
Dew point	46	0	18	+0.197
Relative humidity	24	23	17	+0.177
Heat index	58	1	5	+0.373
Wet-bulb temperature	57	0	7	+0.165
WBGT (shade)	58	0	6	+0.172

469 4.5. Projected 2025–2026 outlook and zonal synthesis

470 Consistent with the modest observed slopes, the projected year-over-year
 471 change was small: the heat index rose by a median of $+0.038^{\circ}\text{C}$ per year
 472 across the 64 districts (range -0.034 to $+0.096^{\circ}\text{C}$ per year), the negative
 473 values corresponding to the districts whose 1980–2024 mean-temperature
 474 trend is itself slightly negative. The zonal synthesis (Table 4) places the
 475 highest projected annual-mean heat index in the Southwest Hot–Dry Zone
 476 (31.0°C) and the Coastal Humid Heat Zone (30.6°C), and the lowest in
 477 the Haor/Wetland Zone (28.0°C); sun WBGT peaked in the coastal belt
 478 (25.5°C). The two continuous-burden conditions reinforce this ordering:

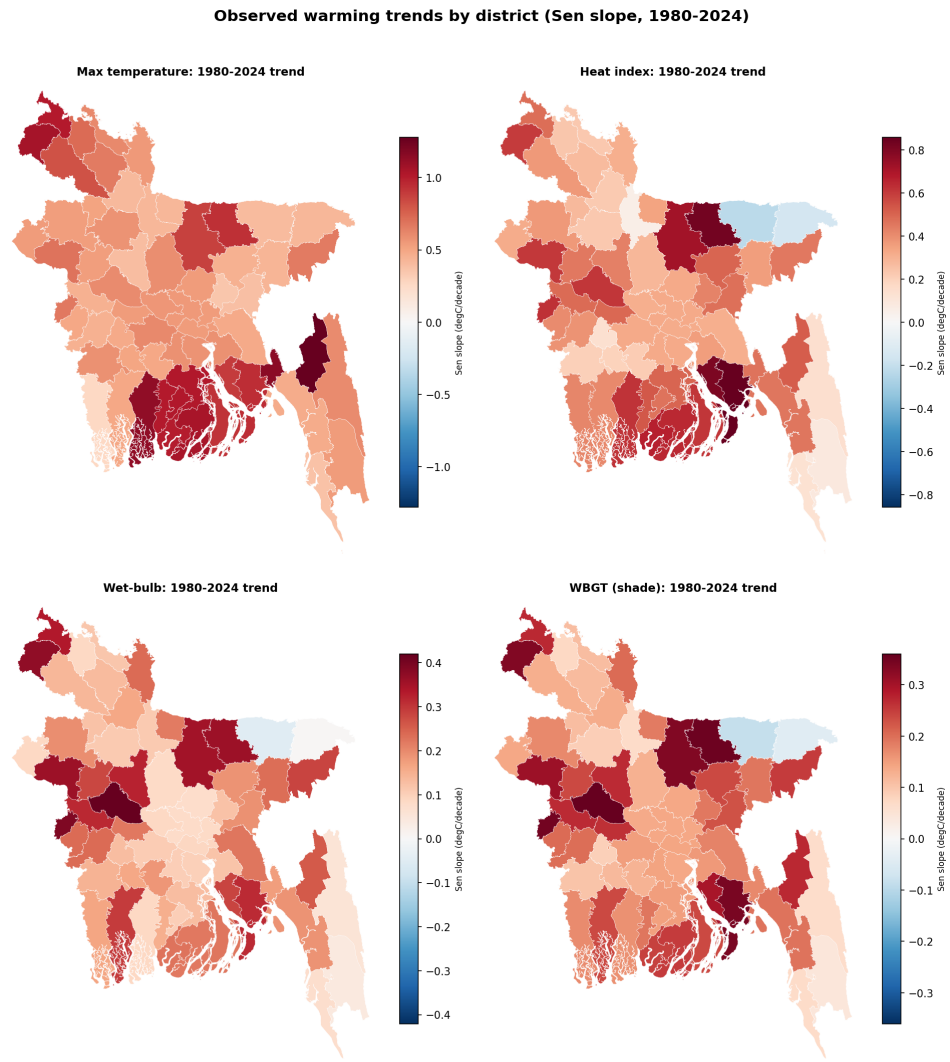


Figure 2: Observed 1980–2024 warming by district (Theil–Sen slope, °C per decade) for maximum temperature, heat index, wet-bulb temperature, and shade WBGT. Red indicates warming, blue cooling. Values are computed after masking missing-as-zero early-record days (Section 3.12).

479 projected cooling degree-days—a proxy for building cooling-energy demand—
 480 were highest in the Southwest Hot–Dry and Coastal Humid Heat zones
 481 (9.1 and 9.0 °C d d^{−1}) and lowest in the Haor/Wetland Zone (7.0), while
 482 projected wet-bulb temperature, the thermodynamic ceiling on evaporative
 483 cooling, followed the same spatial pattern, peaking in the Coastal (23.7 °C)
 484 and Southwest (23.6 °C) zones and again reaching its minimum in the Haor
 485 basin (22.6 °C). The coastal belt therefore emerges as the zone of great-
 486 est combined perceived-heat, humid-heat, and cooling-energy burden. Fig-
 487 ure 6 maps all six continuous conditions at district resolution, confirming
 488 the coastal-south concentration of heat, humidity, and cooling demand. Fig-
 489 ure 3 illustrates the full record–validation–projection narrative for the Dhaka
 490 exemplar, and Figure 4 summarises the projected monthly climatology of all
 491 six key conditions across the six zones, exposing their distinct seasonal sig-
 492 natures. Figure 5 shows the district spread within the hottest (Southwest)
 493 zone.

Table 4: Projected annual-mean thermal indices by climatic zone (zone mean, with the across-district minimum and maximum in parentheses). Values in °C except CDD (°C d d^{−1}).

Zone	Heat index	WBGT (sun)	Wet-bulb	CDD
1. Northwest Extreme Heat	29.6 (28.4–30.7)	24.7	22.9	8.3
2. Central Urban Heat	29.6 (29.3–30.9)	24.8	22.8	8.8
3. Southwest Hot–Dry	31.0 (30.6–31.9)	25.5	23.6	9.1
4. Coastal Humid Heat	30.6 (29.5–31.5)	25.5	23.7	9.0
5. Haor/Wetland Humid	28.0 (26.0–29.5)	24.3	22.6	7.0
6. Hill Tract	29.9 (29.3–30.3)	25.4	23.5	8.6

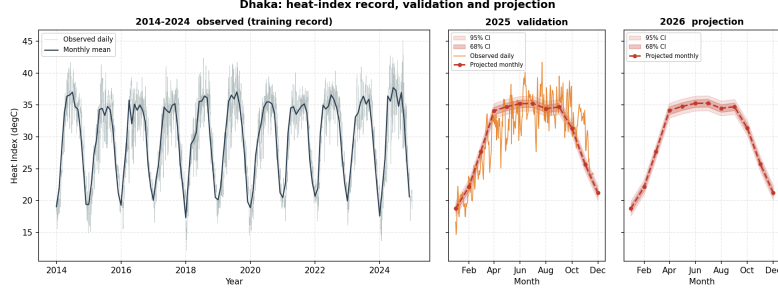


Figure 3: Heat-index record, validation, and projection for the Dhaka exemplar: (left) observed daily heat index 2014–2024 with the monthly mean; (centre) the 2025 validation year, observed daily versus the monthly projection with 68% and 95% confidence bands; (right) the 2026 monthly projection with confidence cloud.

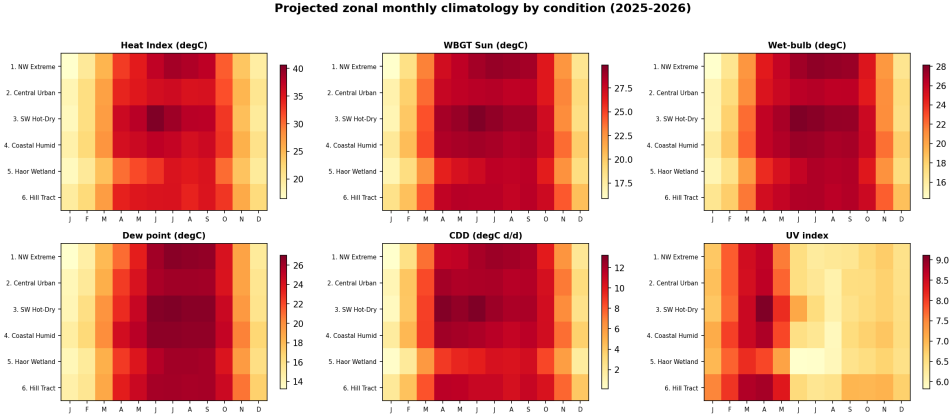


Figure 4: Projected zonal monthly climatology for six heat-stress conditions (heat index, sun WBGT, wet-bulb temperature, dew point, cooling degree-days, and UV index). The temperature-driven conditions peak in the pre-monsoon to monsoon window, whereas the UV index peaks in the dry pre-monsoon (March–April) and falls during the cloudy monsoon — a distinct, radiation-driven seasonality.

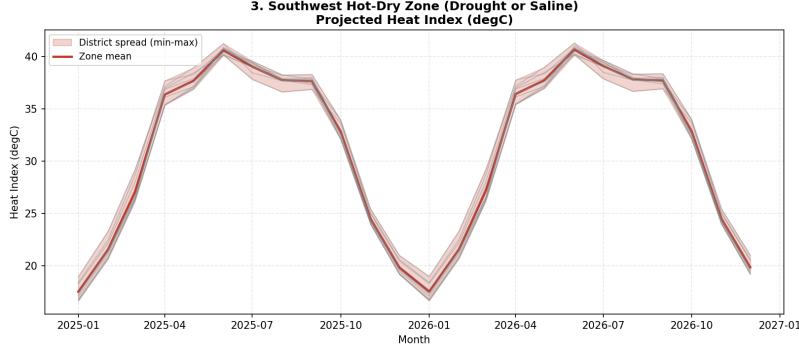


Figure 5: Projected heat index for the Southwest Hot-Dry Zone: the zone mean and the min-max envelope across member districts, illustrating within-zone spread.

4.6. Operational risk categories

Mapping the projected fields onto their risk-category systems yields a policy-facing summary of the seven conditions (Table 5). Under monthly-mean heat index, projected months split between Comfortable (37.8%) and Extreme Caution (51.8%), with the intermediate Caution class at 10.2% and the Danger class essentially absent (0.1%)—the latter a direct consequence of categorising monthly means rather than daily peaks. Outdoor WBGT was predominantly Low (59.0%) to Moderate (37.0%), reaching the High class in 3.9% of months, while indoor (shade) WBGT rarely left the Low class (79.8%). The UV signal was the most consistently hazardous condition: 75.5% of projected months fall in the WHO High class and a further 20.4% in Very High. Dew-point moisture stress was also severe, with 45.5% of months in the Extremely Oppressive class. Across zones, the Southwest Hot-Dry Zone carried the largest Extreme-Caution share for heat index (57.5%) and the only appearance of the Danger class (0.8%), whereas the Haor/Wetland Zone was the mildest (35.4% Extreme Caution); Figure 7 shows the full zonal composition and Figure 8 maps, for each dis-

Projected 2025-2026 annual-mean heat-stress conditions by district

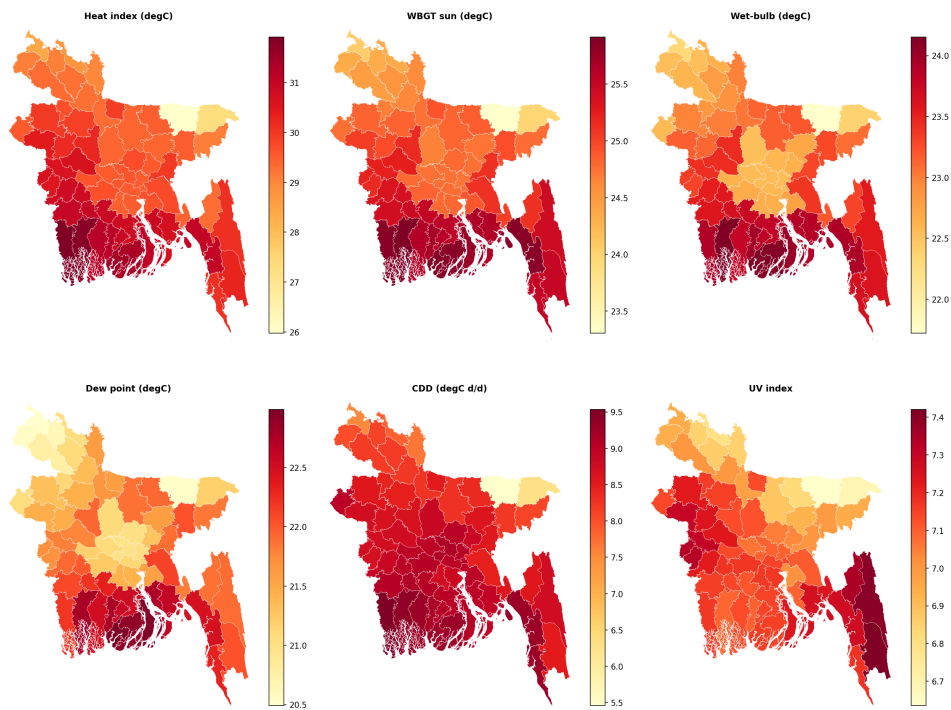


Figure 6: Projected 2025–2026 annual-mean heat-stress conditions by district: heat index, sun WBGT, wet-bulb temperature, dew point, cooling degree-days, and UV index. Every condition is shown at full district resolution rather than aggregated to zones.

511 trict, the number of months per year spent at or above each condition’s
 512 hazard threshold. The months-per-year view resolves the strong spatial and
 513 seasonal gradients that a dominant-class map obscures: the coastal south
 514 endures dew-point levels in the Extremely Oppressive class for roughly seven
 515 to eight months per year against three to four in the northwest, and very-
 516 high UV persists longest in the south.

517 Critically, the categorical layer validated well against observed 2025 (Ta-
 518 ble 6): exact-class agreement ranged from 81.5% (UV) to 89.8% (heat-index
 519 comfort), and agreement to within one adjacent class reached 100% for all
 520 five categorised conditions. The projection therefore reproduces not only
 521 the continuous fields but also their operational risk classification.

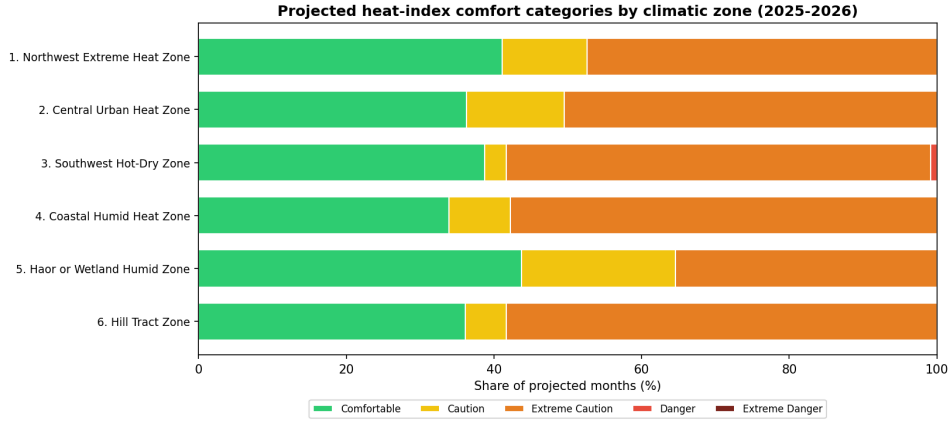


Figure 7: Projected heat-index comfort categories by climatic zone (share of projected months, 2025–2026). The Southwest Hot–Dry Zone carries the largest Extreme-Caution share and the only Danger occurrences; the Haor/Wetland Zone is mildest.

522 4.7. Projection uncertainty

523 The first-order error propagation yielded confidence bands that are ap-
 524 proximately constant across the 24-month horizon, reflecting the dominance

Table 5: Projected risk-category distribution (share of projected district-months, 2025–2026, 64 districts). Only populated categories are shown.

Condition	Category	Share (%)
Heat-index comfort	Comfortable	37.8
	Caution	10.2
	Extreme Caution	51.8
	Danger	0.1
WBGT (sun, outdoor)	Low	59.0
	Moderate	37.0
	High	3.9
WBGT (shade, indoor)	Low	79.8
	Moderate	20.2
UV (WHO)	Moderate	4.2
	High	75.5
	Very High	20.4
Dew-point moisture stress	Comfortable	19.7
	Slightly Humid	6.6
	Humid	16.3
	Very Humid / Oppressive	11.9
	Extremely Oppressive	45.5

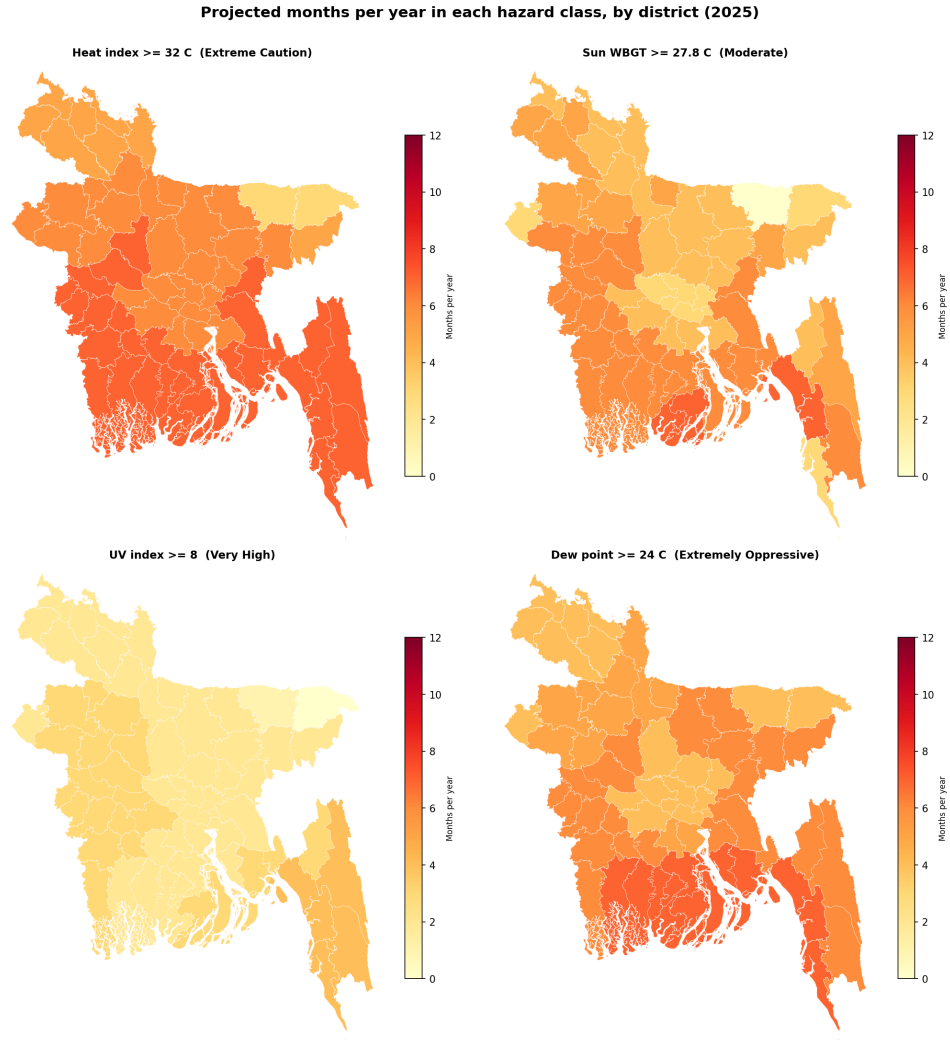


Figure 8: Projected number of months per year (of 12) that each district spends at or above a hazard threshold, for heat index ($\geq 32^{\circ}\text{C}$, Extreme Caution), sun WBGT ($\geq 27.8^{\circ}\text{C}$, Moderate), UV (≥ 8 , Very High), and dew point ($\geq 24^{\circ}\text{C}$, Extremely Oppressive). This months-per-year measure preserves the seasonal and spatial structure that a single dominant-class map collapses.

Table 6: Categorical validation against observed 2025: exact-class agreement and agreement to within one adjacent class, pooled over district-months.

Condition	n	Exact-class (%)	Within one class (%)
Heat-index comfort	704	89.8	100.0
WBGT (sun, outdoor)	704	84.1	100.0
WBGT (shade, indoor)	704	88.8	100.0
UV (WHO)	704	81.5	100.0
Dew-point moisture stress	704	87.1	100.0

of the seasonal model’s error over the small trend extrapolation. For the Dhaka exemplar, the 95% heat-index band was $\pm 1.17^\circ\text{C}$, the wet-bulb band $\pm 0.99^\circ\text{C}$, and the sun-WBGT band $\pm 0.76^\circ\text{C}$. These bands exceed the projected inter-annual trend signal by more than an order of magnitude, and this relationship is stated explicitly to caution against over-interpreting small year-to-year differences in the projection.

5. Discussion

5.1. Principal findings

This study delivers district-resolved, uncertainty-aware near-term projections of seven temperature-correlated heat-stress conditions for all 64 districts of Bangladesh, together with a leakage-free external validation against independent 2025 observations. The recomputed thermal indices validated strongly—heat index, wet-bulb temperature, and WBGT all at median $R^2 \geq 0.95$ —and, when mapped onto their operational risk classes, reproduced the observed 2025 class in 81–90% of district-months exactly and in 100% to within one adjacent class. Because the indices are computed

541 from the projected primary variables rather than predicted directly, this
 542 accuracy is achieved without target leakage and while preserving the phys-
 543 ical coupling among temperature, humidity, and radiation. This validated,
 544 multi-index projection extends prior Bangladeshi heat-projection work that
 545 has largely treated single indices in isolation—most notably the heat-index
 546 appraisal of Rahman et al. [20] and the WBGT trend analysis of Kamal
 547 et al. [9]—and complements single-algorithm forecasting precedents such as
 548 Binte Ahmed et al. [36] and Islam et al. [33] by covering all seven conditions
 549 jointly and validating them out-of-sample.

550 5.2. Skill must be interpreted correctly

551 A central and deliberately transparent result is that, although the ensem-
 552 bles comfortably outperform a seasonal-naive baseline, they are essentially
 553 at parity with monthly climatology (positive skill in only $\sim 32\%$ of district-
 554 variable cases). This is expected rather than disappointing: at monthly
 555 resolution the tropical seasonal cycle dominates the variance, and any com-
 556 petent learner reconstructs the climatological annual march that climatology
 557 also encodes. The contribution of the framework is therefore not a reduc-
 558 tion in monthly error but the provision of a spatially complete, uncertainty-
 559 quantified, and trend-consistent projection—including physiologically rele-
 560 vant indices and their operational risk classes—that a bare climatology can-
 561 not supply. We state this explicitly to discourage over-claiming predictive
 562 skill that the data do not support. This transparency deliberately contrasts
 563 with the common practice of reporting high absolute accuracy—for instance
 564 the R^2 values near 0.99 and 0.88–0.97 reported for Bangladeshi heat-index
 565 and temperature prediction by Binte Ahmed et al. [36] and Islam et al.
 566 [33]—without benchmarking against a climatological reference, which can

567 make routine seasonal reconstruction appear as genuine forecasting skill.

568 *5.3. Physical and spatial interpretation*

569 The observational analysis reveals a coherent but spatially structured
570 warming signal. Maximum temperature increased in every district (median
571 $+0.56^{\circ}\text{C}$ per decade) while minimum temperature remained essentially flat
572 (median $+0.02^{\circ}\text{C}$ per decade), implying a widening diurnal temperature
573 range across much of the country—a pattern with distinct implications for
574 daytime heat exposure and nighttime physiological recovery. Humid-heat
575 indices (heat index, wet-bulb, WBGT) rose in 57–58 of 64 districts, con-
576 firming that the warming is not purely a dry-bulb phenomenon. Spatially,
577 the Southwest Hot–Dry and Coastal Humid Heat zones carry the greatest
578 projected burden across perceived heat, humid heat, and cooling-energy de-
579 mand, whereas the elevated Haor/Wetland basin is consistently mildest; the
580 coastal belt in particular combines the highest wet-bulb temperatures and
581 cooling degree-days, marking it as a priority region. These patterns matter
582 physiologically: the elevated coastal wet-bulb burden erodes the evaporative
583 margin that Raymond et al. [6] identified as the hard limit of human heat
584 tolerance, in the same coastal belt where Ekra et al. [21] found the steepest
585 historical rise in heat discomfort. The widening diurnal range is also epi-
586 demologically salient, since reduced nighttime cooling impairs physiological
587 recovery and compounds the heatstroke-mortality and labour-capacity risks
588 documented for Bangladesh and for outdoor workers more broadly [26, 57].

589 *5.4. Operational risk categories and their message*

590 Translating the projections into established risk classes sharpens the
591 public-health narrative. The UV signal is the most uniformly hazardous

condition: essentially the entire country falls in the WHO High class or above throughout the projection, and dew-point moisture stress reaches the Extremely Oppressive class in almost half of all projected months. Heat-index comfort is dominated by the Extreme Caution class in the warm season, while the Danger class is essentially absent—a direct and important consequence of categorising monthly means rather than daily maxima (Section 3.8), which the categories must be read against. The nationwide reach of the UV and humidity hazards, contrasted with the more spatially variable heat-index and WBGT burden, argues for differentiated adaptation responses: the near-universal High-to-Very-High UV exposure calls for the sun-protection measures codified in the WHO UV-index guidance [17], whereas the humid-heat burden is better addressed through the WBGT-based occupational-safety framework of ISO 7243 [4, 14], whose relevance is amplified by the large humid-heat labour losses estimated for South Asian outdoor workers [56].

5.5. Methodological reflections

Three design choices proved consequential. First, recomputing indices from projected primary variables (rather than modelling them directly) both prevents target leakage and guarantees internal physical consistency of the projected fields. Second, the detrend–residual–retrend construction resolves a failure mode of tree ensembles—their inability to extrapolate a calendar-year feature—so that projected years evolve with the observed climate trend instead of collapsing onto the last training year; imposing the robust 1980–2024 Sen slope additionally keeps the projection consistent with the manuscript’s own trend analysis by construction. Third, the district choropleths exposed a data-quality problem (early-record missing values en-

618 coded as zero) that zonal-median summaries had masked, underscoring the
619 value of full-resolution spatial diagnostics and prompting the physical-floor
620 correction described in Section 3.12.

621 5.6. Limitations

622 Several limitations bound the interpretation. (i) The projection horizon
623 is deliberately restricted to 24 months; because the re-added trend is small
624 relative to the projection uncertainty (the 95% heat-index band exceeds the
625 annual trend signal by more than an order of magnitude), longer horizons
626 would add deterministic drift without validatable information. (ii) Work-
627 ing at monthly-mean resolution necessarily compresses daily extremes, so
628 the risk categories describe typical monthly conditions and systematically
629 under-represent short-duration dangerous events; a daily-resolution exten-
630 sion is the natural next step. (iii) A composite heatstroke-hazard index was
631 implemented but excluded from predictive results for negative out-of-sample
632 skill, and is reported only descriptively. (iv) All meteorological inputs de-
633 rive from a single commercial reanalysis-blended source; despite strong 2025
634 validation, independent station-based corroboration would strengthen con-
635 fidence, and the early-record inhomogeneity in a subset of districts (miti-
636 gated here by masking) illustrates the sensitivity of long-run trends to input
637 quality. (v) The models add little skill over climatology at monthly scale,
638 as discussed above. The 24-month horizon is complementary to, rather
639 than competitive with, the multi-decadal CMIP6 projections available for
640 Bangladesh and the wider region [58, 44]: those studies bound the long-term
641 trajectory under emission scenarios, whereas the present framework supplies
642 the near-term, district-resolved, externally validated detail that dynamical
643 downscaling cannot yet deliver at this resolution.

644 5.7. Outlook

645 Several extensions follow naturally. A daily-resolution implementation
646 would recover the short-duration extremes that monthly means smooth away
647 and let the risk categories reflect daily peaks—the regime in which the heat-
648 stroke indicator excluded here may regain skill [49]. Coupling the projected
649 fields with health-surveillance and cooling-energy records would convert the
650 physical outlook into quantified public-health and infrastructure burdens
651 [26]. Methodologically, the framework can absorb recent advances in the in-
652 dividual conditions—reanalysis-based UV retrieval [51] and attention-based
653 dew-point models with explainability [55]—and its near-term projections
654 can be systematically benchmarked against CMIP6 dynamical downscaling
655 [58] to bridge near-term operational forecasting and long-term climate pro-
656 jection.

657 6. Conclusion

658 This study introduced the Temperature-Related Conditions (TRC)
659 framework and applied it to project seven physiologically and operationally
660 relevant heat-stress conditions across all 64 districts of Bangladesh. By
661 training gradient-boosting ensembles on the 2014–2024 record with future-
662 known seasonal features, re-imposing the observed 1980–2024 trend, and
663 recomputing every thermal index from the projected primary variables,
664 the framework produced district-level 24-month projections that validated
665 strongly against independent 2025 observations—median R^2 of 0.95–0.98
666 for the key indices, with operational risk classes reproduced to within one
667 category in 100% of district-months.

668 The observational analysis revealed a coherent but spatially structured

warming signal: maximum temperature rose in every district (median +0.56 °C per decade) while minimum temperature remained essentially flat, implying a widening diurnal temperature range, and the humid-heat indices increased across most of the country. Spatially, the coastal and southwestern zones emerged as the areas of greatest combined perceived-heat, humid-heat, and cooling-energy burden, whereas ultraviolet and oppressive-humidity hazards were effectively nationwide. Crucially, the ensembles were found to be at parity with monthly climatology; the framework’s contribution is therefore not improved monthly accuracy but a spatially complete, uncertainty-aware, and trend-consistent outlook—delivered at full district resolution—that supports targeted heat-adaptation planning. One indicator, heatstroke risk, was excluded from the predictive results for a lack of out-of-sample skill and reported only descriptively.

Several directions follow naturally. Extending the analysis to daily resolution would capture the short-duration extremes that monthly means smooth away and would allow the risk categories to reflect daily peaks rather than typical monthly conditions. Coupling the projected indices with health-surveillance and energy-demand records would translate this physical outlook into quantified public-health and infrastructure burdens [**verify-intended**]. More broadly, the leakage-free, trend-consistent, and externally validated design demonstrated here offers a transferable template for district-level heat-stress projection in other data-constrained, climatically heterogeneous regions.

CRedit authorship contribution statement

[**PLACEHOLDER: CRediT roles per author.**]

694 **Declaration of competing interest**

695 **[PLACEHOLDER: competing-interest statement.]**

696 **Data availability**

697 Daily meteorological data are available from the Visual Crossing Weather
698 API (<https://www.visualcrossing.com>). **[PLACEHOLDER: final data-**
699 **availability wording and any access/licence details.]**

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